MTH401 - Differential Equations (Quiz#3)

Question # 1 of 10 (Start time: 12:08:53 PM, 03 February 2023)

Legendre polynomials are normal with respect to the weight function

Select the correct option

- $w(x) = \frac{n+1}{2n}$
- $w(x) = \frac{2n^2+1}{n}$
- $w(x) = \frac{2n+1}{n}$
- $w(x) = \frac{2n+1}{n^2}$

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is an other solution
origin, this combination represents the general
Note: The method of Frobenius may not always provide 2 solutions.
tial equation $xy'' + 3y' - y = 0$ has regular singular

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Question # 7 of 10 (Start time: 08:47:35 PM, 03 February 2023)

x=0 is a(an) ------- point of the differential equation $y'' + xy' + 2y = 0$.

Select the correct option

- Regular singular
- Ordinary
- Irregular singular
- Singular



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.1 Regular and Irregular Singular Points

Singular point $x = x_0$ of the given equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ is said to

egular singular point if both $(x - x_0)P(x)$ and $(x - x_0)^2 Q(x)$ are analytic at x_0. A gular point that is not regular is said to be an *irregular singular point* of the equation.

1.1 Polynomial Coefficients

he coefficients in the given differential equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ are ynomials with no common factors, above definition is equivalent to the following:

Let $a_2(x_0) = 0$ *Form* $P(x)$ *and* $Q(x)$ *by reducing* $\frac{a_1(x)}{a_2'(x)}$ *and* $\frac{a_0(x)}{a_2(x)}$ *to lowest terms, respectively. If the factor* $(x - x_0)$ *appears at most to the first powers in the denominator of* $P(x)$ *and at most to the second power in the denominator of* $Q(x)$, *then* $x = x_0$ *is a* ***regular singular point.***

1 $x = \pm 2$ are singular points of the equation $(x^2 - 4)^2 y'' + (x - 2)y' + y = 0$

, we find that

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Question # 6 of 10 (Start time: 05:16:30 PM, 03 February 2023)

To solve a differential equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$$

about a regular singular point we employ the _____ theorem.

Select the correct option

○	Frobenius
○	Bessel
○	Legendre
○	none of them

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$$\frac{dx_n}{dt} = g_n\left(t, x_1, x_2, \ldots, x_n\right)$$

is also particularly important in advanced mathematics. This system of n first-order equations is called and ***n*th-order system.**

2. Every nth-order differential equation

$$y^{(n)} = F\left(t, y, y', \ldots, y^{(n-1)}\right)$$

as well as ***most systems*** of differential equations, can be reduced to the ***nth-order system.***

37.2 Linear Normal Form

A particularly, but important, case of the *nth-order system* is of those systems having the linear normal or canonical form:

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34 Legendre's Differential Equation

A second order linear differential equation of the form

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

is called Legendre's differential equation and any of its solution is called Legendre's function. If n is positive integer then the solution of Legendre's differential equation is called a Legendere's polynomial of degree n and is denoted by $P_n(x)$.

We assume a solution of the form $y = \sum_{k=0}^{\infty} C_k x_k$

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33 Bessel's Differential Equation

A second order linear differential equation of the form $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+(x^2-\nu^2)y=0$ is called Bessel's differential equation.

Solution of this equation is usually denoted by $J_\nu(x)$ and is known as Bessel's function. This equation occurs frequently in advanced studies in applied mathematics, physics and engineering.

33.1 Series Solution of Bessel's Differential Equation

Bessel's differential equation is $x^2y''+xy'+(x^2-\nu^2)y=0$ (1)

If we assume that $y=\sum_{n=0}^{\infty}C_nx^{n+r} \Rightarrow y'=\sum_{n=0}^{\infty}C_n(n+r)x^{n+r-1} \Rightarrow$

$$y''=\sum_{n=0}^{\infty}C_n(n+r)(n+r-1)x^{n+r-2}$$

So that

The characteristic equation of 1st order homogeneous differential equation

$$\begin{bmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Select the correct option

Reload Math Equations

- $del(A + \lambda I) = \begin{vmatrix} 1+\lambda & 2 \\ 3 & 4+\lambda \end{vmatrix}$
- $del(A - \lambda I) = \begin{vmatrix} 1-\lambda & 2 \\ 3 & 4-\lambda \end{vmatrix}$
- $del(X - \lambda I) = \begin{vmatrix} x-\lambda & 2 \\ 3 & y-\lambda \end{vmatrix}$

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odd 1/1

Question # 6 of 10 (Start time: 01:06:01 AM, 01 February 2023)

Total Marks: 1

Bessel functions of index half an _____ integer are called Spherical Bessel functions.

Select the correct option

- (●) odd
- ( ) even

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Question # 7 of 10 (Start time: 08:31:40 AM, 01 February 2023) Total Marks: 1

The differential equation

$$5\frac{dx}{dt} + 5x - \frac{dy}{dt} = e^{2t}$$

in terms of differential operator is _____.

Select the correct option

Reload Math Equations

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- (●) $(D+1)5x - Dy = e^{2t}$
- ( ) $(D+1)5x + Dy = e^{2t}$
- ( ) $(D-1)5x - Dy = e^{2t}$
- ( ) $(D+1)5x - Dy = e^{t}$

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Question # 4 of 10 (Start time: 08:27:34 AM, 01 February 2023) **Total Marks:**

A second order linear differential equation of the form

$$(1-x^2)y - 2xy' + n(n+1)y = 0$$

is called _____ differential equation.

Select the correct option

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- ○ Picard
- ○ Bernoulli
- ○ Bessel
- ◉ Legendre

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34 Legendre's Differential Equation

A second order linear differential equation of the form

$$\left(1-x^2\right)y''-2xy'+n(n+1)y=0$$

is called Legendre's differential equation and any of its solution is called Legendre's function. If n is positive integer then the solution of Legendre's differential equation is called a Legendere's polynomial of degree n and is denoted by $P_n(x)$.

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Question # 5 of 10 (Start time: 07:45:56 AM, 01 February 2023) **Total Marks:**

The horizontal lines in a matrix are called the _____ of the matrix.

Select the correct option

- (●) rows
- ( ) columns

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Q. Spherical Bessel function are used in many physical problems.

False

True

292

Q. If $X_1 = \begin{pmatrix} e^{-2t} \\ -e^{-2t} \end{pmatrix}$, then $\frac{dx_1}{dt} = ?$

$\begin{pmatrix} -2e^{-2t} \\ -2e^{-2t} \end{pmatrix}$ $\begin{pmatrix} 2e^{-2t} \\ -2e^{-2t} \end{pmatrix}$ $\begin{pmatrix} 2e^{-2t} \\ 2e^{-2t} \end{pmatrix}$ $\begin{pmatrix} -2e^{-2t} \\ 2e^{-2t} \end{pmatrix}$

365

Q. The horizontal line in a matrix are called the ___ of the matrix.

Columns

rows

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Question # 3 of 10 (Start time: 10:51:25 AM, 01 February 2023)

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The vertical lines in a matrix are called the _____ of the matrix.

Select the correct option

- rows
- columns

Question # 2 of 10 (Start time: 01:13:48 AM, 01 February 2023) Total Marks:

$$If\ K = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ is an eigenvector of the given matrix A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} then\ AK\ is$$

Select the correct option

Reload Math Equations

○	$-1K$
○	-1
○	1
◉	$1K$

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(b) The singular points need not be real numbers.

The equation $(x^2+1)y''+2xy'+6y=0$ has the singular points at the solutions of $x^2+1=0$, namely, $x=\pm i$.

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$$4\frac{d^3y}{dt^3} + y = e^t$$

$$y = x_1, y' = x_1', y'' = x_2' \text{and } y''' = x_3'$$

$$x_1' = x_2, x_2' = x_3, x_3' = \frac{1}{4}x_1 - \frac{1}{4}e^t$$

$$x_1' = x_2, x_2' = x_3, x_3' = \frac{-1}{4}x_1 + \frac{1}{4}e^t$$

$$x_1' = x_2, x_2' = x_3, x_3' = \frac{1}{4}x_1 + \frac{1}{4}e^t$$

$$x_1' = x_2, x_2' = x_3, x_3' = -\frac{1}{4}x_1 - \frac{1}{4}e^t$$

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31.2 Method of Frobenius

To solve a differential equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ about a regular singular point we employ the Frobenius' Theorem.

31.2.1 Frobenius' Theorem

If $x = x_0$ is a regular singular point of equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$, then there exists at least one series solution of the form

$$y = (x - x_0)^r \sum_{n=0}^{\infty} c_n (x - x_0)^n = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r}$$

where the number r is a constant that must be determined. The series will converge

$0 < x - x_0 < R.$

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Eigenvalues of the following homogeneous system of Differential equation

$$\frac{dx}{dt} = x + y, \frac{dy}{dt} = -y$$

with coefficient matrix

$$\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$$

Select the correct option

Reload Math Equations

- ○ $\lambda = 1, 1$
- ■ $\lambda = 1, -1$
- ○ None of them
- ○ $\lambda = -1, -1$

The linear normal form of

$$4\frac{d^3y}{dt^3} + y = e^t$$

A) $x_1' = x_2$, $x_2' = x_3$, $x_3' = \frac{1}{4}x_1 + \frac{1}{4}e^t$

B) $x_1' = x_2$, $x_2' = x_3$, $x_3' = -\frac{1}{4}x_1 + \frac{1}{4}e^t$

C) $x_1' = x_2$, $x_2' = x_3$, $x_3' = \frac{1}{4}x_1 - \frac{1}{4}e^t$

D) $x_1' = x_2$, $x_2' = x_3$, $x_3' = -\frac{1}{4}x_1 - \frac{1}{4}e^t$

$y = x_1$, $y' = x_2$, $y'' = x_3$ ... $y^{n-1} = x_n$

$y' = x_1' = x_2$, $y'' = x_2' = x_3$, $y''' = x_3' = x_4$ ... $y^{n-1} = x_{n-1}' = x_n$, $y^{(n)} = x_n'$.

4) $y''' = -y + e^t$

$y''' = -\frac{1}{4}y + \frac{1}{4}e^t$

$x_3' = -\frac{1}{4}x_1 + \frac{1}{4}e^t$

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MTH401 - Differential Equations (Quiz#3)

Quiz Start Time: 12:19 PM, 03 February 2023

Question # 4 of 10 (Start time: 12:22:54 PM, 03 February 2023)

Total Marks:

The equation, $(x^2-4)^2\frac{d^2y}{dx^2}+(x-2)\frac{dy}{dx}+y=0$, has the singular point(s) at $x=$ _____.

Select the correct option

Reload Math Equations

○	± 2
○	$\pm i$
○	± 1
○	0

Click to Save Answer & Move to Next Question

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Question # 7 of 10 (Start time: 01:06:48 AM, 01 February 2023)

Total Marks: 1

It is NOT always possible to solve the given system for the highest-order derivative of each dependent variable.

Select the correct option

- (●) True
- ( ) False



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Question # 5 of 10 (Start time: 01:05:06 AM, 01 February 2023) Total Mark

Vectors $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} -66 \\ 99 \end{pmatrix}$ are linearly ------.

Select the correct option

Reload Math Equation

- Independent
- Dependent



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Click to Save Answer & Move to Next Question

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Question # 6 of 10 (Start time: 12:02:35 PM, 03 February 2023)

Operator method is based on process of

Select the correct option

○	None of these.
○	Addition of dependent variables.
○	Elimination of dependent variables.
○	Obtaining new dependent variables.

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MTH401 - Differential Equations (Quiz#3)

Quiz Start Time: 12:19 PM, 03 February 2023

Question # 9 of 10 (Start time: 12:27:58 PM, 03 February 2023)

Total Marks:

The differential equation

$$5\frac{dx}{dt} + 5x - \frac{dy}{dt} = e^{2t}$$

in terms of differential operator is _____.

Select the correct option

Reload Math Equations

○	$(D+1)5x + Dy = e^{2t}$
○	$(D+1)5x - Dy = e^{t}$
○	$(D-1)5x - Dy = e^{2t}$
○	$(D+1)5x - Dy = e^{2t}$

Click to Save Answer & Move to Next Question

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Question # 2 of 10 (Start time: 12:20:43 PM, 03 February 2023) **Total Marks:**

The regular singular point of the differential equation

$$(x^2 - 4)^2 y + (x - 2)y' + y = 0$$

is _____

Select the correct option

Reload Math Equations

- ○ 2
- ○ 1
- ○ -2
- ○ -1

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estion # 2 of 10 (start time: 12:31:40 PM, 03 February 2023)

Operator method or systematic elimination method of solution of a system of linear homogeneous or linear non-homogeneo equations provides us _____ differential equation in one of the dependent variables that has not been eliminated

lect the correct option

- many
- single



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$$A = \begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}$$

Clear

Select the correct option

Reload Math Equations

○	$\lambda_1 = 5, \lambda_2 = 5$
○	$\lambda_1 = 5, \lambda_2 = -5$
○	$\lambda_1 = 3, \lambda_2 = 7$
○	$\lambda_1 = -5, \lambda_2 = -5$

Click to Save Answer & Move to Next Question

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MTH401 Quiz. No.3 Lec#34 to41

Q. The Eigenvalues of a matrix are given by the roots of characteristic Equation

351

True False

Q. $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, k is a constant then

353

$A=kA$ $AK=KA$ $-AK=AK$ $AK \neq KA$

Q. Operator method is based on Process of

Elimination of dependent variables Addition of dependent variables

obtaining new dependent variables None of these 305

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MTH 401 Quiz #3 Lec # 34 to 41

Q. The Simultaneous ordinary differential equation involves two or more equations that contain derivaties of two or more unknown functions of ——— independent variable.

2 2 infinite.

(1)

304

Q. Legendre Polynomials are normal with respect to the weight function.

$w(x) = \frac{2n+1}{n}$ $w(x) = \frac{n+1}{2n}$ $w(x) = \frac{2n^2+1}{n}$ $w(x) = \frac{2n+1}{n^2}$

303

Q. If n is positive integer, then the Solution of legendre's differential equation is.

Called a Legendre's ——— of degree n.

294

function

Polynomial.

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Question # 6 of 10 (Start time: 08:43:36 AM, 01 February 2023) **Total Marks:** 1

The simultaneous ordinary differential equations involve two or more equations that contain derivatives of two or more unknown functions of _____ independent variable.

Select the correct option

Reload Math Equations

- 2
- infinite
- 2
- 1

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A second order differential equation of the form

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - v^2)y = 0$$

is called _____ differential equation.

elect the correct option

Reload Math Equations

- Legendre
- Bessel
- Bernoulli
- Picard

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Question # 6 of 10 (Start time: 12:25:00 PM, 03 February 2023)

Total Marks:

A matrix A is defined as

$$A = \begin{bmatrix} a & b \\ & \\ c & d \end{bmatrix}$$, k is a constant then

Select the correct option

Reload Math Equations

- $Ak \neq kA$
- $Ak = kA$
- $-Ak = Ak$
- $A = kA$

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Question # 1 of 10 (Start time: 12:19:18 PM, 03 February 2023)

Total Marks: 1

The linear normal form of

$$2\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 5y = 0$$

by using

$$y = x_1, y' = x_1' \text{and } y'' = x_2'$$

is _____

Select the correct option

Reload Math Equations

○	$x_1' = x_2, x_2' = -2x_2 + \frac{5}{2}x_1$
○	$x_1' = x_2, x_2' = 2x_2 - \frac{5}{2}x_1$
○	$x_1' = x_2, x_2' = -2x_2 - \frac{5}{2}x_1$
○	$x_1' = x_2, x_2' = 2x_2 + \frac{5}{2}x_1$

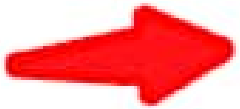
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The Legendre polynomials can also be generated by Rodrigues formula, $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n}(x^2 - 1)^n$.

False

True



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sec(s)

MTH401 - Differential Equations (Quiz#3) Quiz Start Time: 12:19 PM, 03 February 2023

Question # 5 of 10 (Start time: 12:24:00 PM, 03 February 2023) Total Marks: 1

If

$$A = \begin{bmatrix} 1 & 5 & 3 \\ 2 & 4 & 7 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 2 & 1 \\ 3 & 5 & 6 \\ 8 & 7 & 5 \end{bmatrix}$$

then the order of matrix

$$A \times B$$

is

Select the correct option

Reload Math Equations

○	None of them
○	3×2
○	3×3
○	2×3

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If

$$A = \begin{bmatrix} 1 & 5 & 3 \\ 2 & 4 & 7 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 2 & 1 \\ 3 & 5 & 6 \\ 8 & 7 & 5 \end{bmatrix}$$

then the order of matrix

$$A \times B$$

is

Select the correct option

Reload Math Eq

○	2×3
○	3×3
○	None of them
○	3×2

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Question # 7 of 10 (Start time: 12:25:58 PM, 03 February 2023) Total Marks: 1

$$\frac{dA}{dt} = \left[\frac{da_{ij}}{dt}\right]_{m\times n}$$ is the _____ of a matrix of a function.

Select the correct option

Reload Math Equations

- integral
- column
- row
- derivative

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Select the correct option

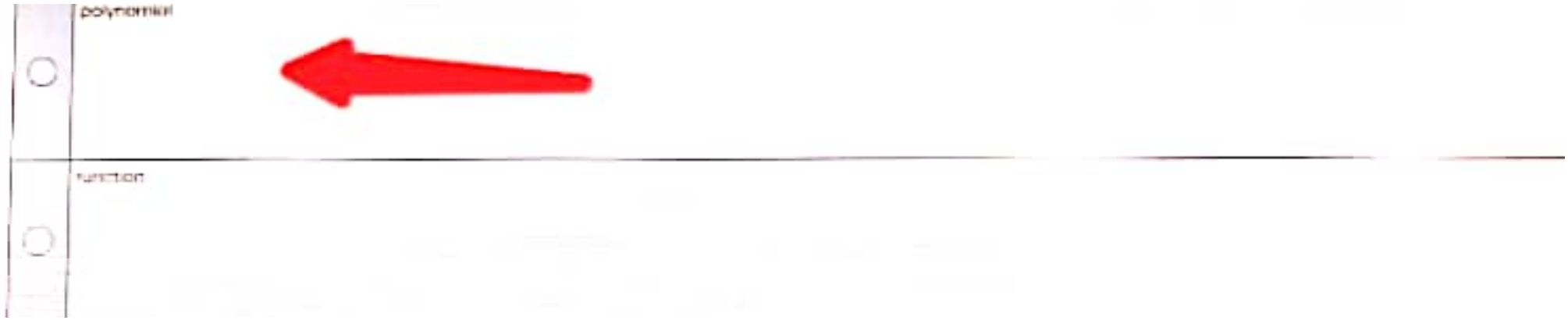
Reload Math Equations

- ○ infinite
- ○ $n+1$
- ○ $n-1$
- ○ n^{th}



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The matrix form of non-homogeneous system of differential equations

$$X' = 3x + 4y - 9z$$
$$X' = 6x - y$$
$$X' = 10x - 4y + 3z$$

$$X' = \begin{bmatrix} 3 & 4 & -9 \\ 6 & -1 & 0 \\ 10 & -4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$X' = \begin{bmatrix} 3 & 4 & -9 \\ 6 & 0 & 0 \\ 4 & 10 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$X' = \begin{bmatrix} 3 & 4 & -9 \\ 6 & -1 & 0 \\ 10 & -4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$X' = \begin{bmatrix} 3 & -9 & -4 \\ 6 & 0 & -1 \\ 10 & 3 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

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Question # 4 of 10 (Start time: 01:10:03 PM, 03 February 2023) Total Marks:

A matrix of the from $B = \begin{bmatrix} b_{11} \\ b_{21} \\ . \\ . \\ . \\ b_{m1} \end{bmatrix}_{m \times 1} = [b_{i1}]_{m \times 1}$ is called

Select the correct option

Reload Math Equations

- column Matrix
- Row matrix

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Question # 3 of 10 (Start time: 01:09:16 PM, 03 February 2023) Total Mark

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} u & v \\ x & y \end{bmatrix}$ then $A + B$ is

Select the correct option

Reload Math Equations

- $A + B = \begin{bmatrix} a+u & b+v \\ c+x & d+y \end{bmatrix}$
- $A + B = \begin{bmatrix} a+u & b+x \\ c+v & d+y \end{bmatrix}$
- $A + B = \begin{bmatrix} au & bx \\ cv & dy \end{bmatrix}$
- $A + B = \begin{bmatrix} au & bv \\ cx & dy \end{bmatrix}$

Click to Save Answer & Move to Next Question

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By converting the equation

$$(x^2 + 4x + 4)y'' + (x - 2)y' + y = 0$$

into general form

$$y'' + P(x)y' + Q(x)y = 0$$

$Q(x)$ is

Select the correct option

Reload M

○	$\frac{1}{(x-2)(x+2)^3}$
○	None of these
○	$\frac{1}{(x-2)(x+2)}$
○	$\frac{1}{(x+2)^2}$

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Operator method or systematic elimination method of solution of a system of linear homogeneous or linear non-homogeneous differential equations provides us _____ differential equation in one of the dependent variables that has not been eliminated

Select the correct option

- ○ single
- ○ many

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estion # 6 of 10 (Start time: 12:36:31 PM, 03 February 2023)

The non - trivial solution of the system of differential equation exist only when

lect the correct option

- $det(A + \lambda I) = 0$
- $det(A - \lambda I) = 0$
- $det(X - \lambda I) = 0$
- $det(A - \lambda I) \neq 0$

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on # 8 of 10 (Start time: 12:59:21 PM, 03 February 2023)

If $K = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ is an eigenvector of the given matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ then AK is

the correct option

Reload Math Equat

-1

$1K$

$-1K$

1

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Question # 6 of 10 (start time: 12:58:22 PM, 03 February 2023)

The second order linear differential equation

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - 49) = 0$$

is a Bessel equation of degree

Select the correct option

Reload Ma

- 3
- none of these
- 2
- 1

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Every n^{th} order differential equation, $y^{(n)} = F\left(t, y, y', \ldots, y^{(n-1)}\right)$, as well as most systems of differential equations, could be reduced to the _____ order system.

Choose the correct option

Reload Math Equa

n^{th}

$n - 1$

$n + 1$

infinite

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MTH401 - Differential Equations (Quiz#3) **Quiz Start Time: 01:07 PM, 03 February 2023**

Question # 10 of 10 (Start time: 01:14:24 PM, 03 February 2023) **Total Marks:**

The second order linear differential equation

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - 49) = 0$$

is a Bessel equation of degree

Select the correct option

Reload Math Equations

- 3
- none of these
- 1
- 2

Click to Save Answer & Move to Next Question

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The equation, $(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 30y = 0$, has the singular point(s) at $x =$ _____.

Select the correct option

Reload Math Equ

- ○ $(I) \quad \pm 1$
- ○ $(III) \quad -i$
- ○ $(II) \quad i$
- ○ (IV) Both (II) and (III).

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It is NOT always possible to solve the given system for the highest-order derivative of each dependent variable

lect the correct option

- True
- False

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$$X_1 = \begin{pmatrix} e^{2t} \\ -e^{2t} \end{pmatrix}$$

$$\frac{d}{dt}(X_1)$$

correct option

$$\begin{pmatrix} 2e^{2t} \\ -2e^{2t} \end{pmatrix}$$

$$\begin{pmatrix} -2e^{2t} \\ 2e^{2t} \end{pmatrix}$$

$$\begin{pmatrix} 2e^{2t} \\ 2e^{2t} \end{pmatrix}$$

$$\begin{pmatrix} -2e^{2t} \\ -2e^{2t} \end{pmatrix}$$



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3)

$\frac{dA}{dt} = \left[\frac{da_{ij}}{dt}\right]_{m\times n}$ is the _____ of a matrix of a function.

integral

row



derivative

column

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Question # 3 of 10 (Start time: 03:21:56 PM, 03 February 2023)

To solve a differential equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$$

about a regular singular point we employ the _____ theorem

Select the correct option

Reload M

○	none of them
○	Legendre
○	Frobenius
○	Bessel

Click to Save Answer & Move to Ne

Scanned with CamScanner

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} u & v \\ x & y \end{bmatrix}$ then $A + B$ is

$$A + B = \begin{bmatrix} au & bv \\ cx & dy \end{bmatrix}$$



$$A + B = \begin{bmatrix} a+u & b+v \\ c+x & d+y \end{bmatrix}$$

$$A + B = \begin{bmatrix} au & bx \\ cv & dy \end{bmatrix}$$

Scanned with CamScanner

A matrix A is defined as

$$A = \begin{bmatrix} a & b \\ \\ c & d \end{bmatrix}$$, k is a constant then

Select the correct option

Reload Math

◉	$-Ak = Ak$
○	$Ak \neq kA$
○	$Ak = kA$
○	$A = kA$

Click to Save Answer & Move to Next

Scanned with CamScanner

The equation, $(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 30y = 0$, has the singular point(s) at $x =$ _____.

he correct option

Reload Math Equ

(I) ± 1



(IV) Both (II) and (III).

(II) i

(III) $-i$

Click to Save Answer & Move to Next Que

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tion # 9 of 10 (Start time: 03:28:10 PM, 03 February 2023) To

The equation, $\det(A - \lambda I) = 0$, is known as the characteristic equation.

t the correct option

Reload Math E

True



False

Click to Save Answer & Move to Next Q

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The given system without the use of matrices

$$\frac{d}{dt}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 5 & 7 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t$$

is _____

Select the correct option

Reload

- ○ $\frac{dx}{dt} = 2x + 4y - e^t; \frac{dy}{dt} = 5x + 7y - e^t$
- ○ $\frac{dx}{dt} = 2x + 4y + e^t; \frac{dy}{dt} = 5x + 7y + e^t$
- ○ $\frac{dx}{dt} = 2x + 4y + e^t; \frac{dy}{dt} = 5x + 7y - e^t$
- ○ $\frac{dx}{dt} = 2x + 4y + e^{-t}; \frac{dy}{dt} = 5x + 7y + e^{-t}$

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$$\frac{dA}{dt} = \left[\frac{da_{ij}}{dt}\right]_{m \times n}$$ is the _____ of a matrix of a function.

Select the correct option

Reload Math

- ○ row
- ○ derivative
- ○ column
- ○ integral

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Question # 2 of 10 (Start time: 03:21:15 PM, 03 February 2023)

The vertical lines in a matrix are called the _____ of the matrix

Select the correct option

- ○ rows
- ○ columns



Click to Save Answer & Move to

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Question # 1 of 10 (Start time: 03:44:22 PM, 03 February 2023)

To solve a differential equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$$

about a regular singular point we employ the _____ theorem

Select the correct option

○	Bessel
○	Legendre
○	Frobenius
○	none of them

Click to S

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a solution of Bessel equation is denoted by _____

t the correct option

Reload Math Equat

$P_n(x)$

$B_n(x)$

$J_n(x)$

$L_n(x)$

Click to Save Answer & Move to Next Quest

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The linear normal form of

$$2\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 5y = 0$$

,by using

$$y = x_1, y' = x_1' \text{and } y'' = x_2'$$

, is_____.

Select the correct option

○	$x_1' = x_2, x_2' = -2x_2 + \frac{5}{2}x_1$
○	$x_1' = x_2, x_2' = 2x_2 + \frac{5}{2}x_1$
○	$x_1' = x_2, x_2' = -2x_2 - \frac{5}{2}x_1$
○	$x_1' = x_2, x_2' = 2x_2 - \frac{5}{2}x_1$

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MTH401 - Differential Equations (Quiz#3)

Question # 2 of 10 (Start time: 03:45:58 PM, 03 February 2023)

Solution of Legendre's differential equation is called

Select the correct option

○	Legendre function.
○	Recurrence relation.
○	Legendre differential.
○	Legendre polynomial.

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PM, 03 February 2023)

Thn non-trivial solution of the system exists only when

$$\det(A - \lambda I) = _____.$$

Select the correct option

- zero
- non-zero



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MTH401 - Differential Equations (Quiz#3)

Question # 4 of 10 (Start time: 03:47:34 PM, 03 February 2023)

Bessel equation occurs frequently in advances studies in _____

Select the correct option

○	applied mathematics
○	physics
○	All of these
○	engineering

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MTH401 - Differential Equations (Quiz#3)

Question # 6 of 10 (Start time: 03:48:49 PM, 03 February 2023)

Operator method is based on process of

Select the correct option

- ○ None of these.
- ○ Addition of dependent variables.
- ○ Obtaining new dependent variables.
- ○ Elimination of dependent variables.



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es :ion # 6 of 10 (Start time: 03:52:19 PM, 03 February 2023)

The horizontal lines in a matrix are called the _____ of the matrix.

elect the correct option

rows



columns

Every n^{th} order differential equation, $y^{(n)} = F\left(t, y, y', \ldots, y^{(n-1)}\right)$, as well as most systems of differential equations, could be reduced to the _____ order system.

:t option

Reload Math Eq

$n+1$

$n-1$



n^{th}

infinite

Click to Save Answer & Move to Next Qu

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1401 - Differential Equations (Quiz#3)

Question # 9 of 10 (Start time: 03:56:10 PM, 03 February 2023)

$x=0$ is a(an) -------- point of the differential equation: $y'' + xy' + 2y = 0.$

Select the correct option

- ○ Regular singular
- ○ Irregular singular
- Ordinary
- Singular

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Let $A = \begin{bmatrix} a & b \\ & \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} u & v \\ & \\ x & y \end{bmatrix}$ then $A + B$ is

$$A + B = \begin{bmatrix} au & bv \\ & \\ cx & dy \end{bmatrix}$$

$$A + B = \begin{bmatrix} au & bx \\ & \\ cv & dy \end{bmatrix}$$

$$A + B = \begin{bmatrix} a + u & b + x \\ & \\ c + v & d + y \end{bmatrix}$$

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Question # 3 of 10 (Start time: 12:47:54 AM, 01 February 2023) Total Marks:

Let matrix

$$A=\begin{bmatrix}1 & 5 & 3\\ 2 & 4 & 7\end{bmatrix} \text{ and } B=\begin{bmatrix}0 & 2 & 1\\ 3 & 5 & 1\end{bmatrix}$$

then A-B is _____.

Select the correct option

Reload Math Equations

- $\begin{bmatrix}1 & 3 & 2\\ 1 & -1 & 6\end{bmatrix}$
- $\begin{bmatrix}1 & 3 & 2\\ 1 & 1 & 6\end{bmatrix}$
- $\begin{bmatrix}1 & 3 & 2\\ -1 & -1 & 6\end{bmatrix}$
- $\begin{bmatrix}1 & 3 & 2\\ -1 & 1 & 6\end{bmatrix}$

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Go to Settings to activate Windows.

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The solution of Legendre differential equation is denoted by _____.

Select the correct option

Reloa

○	$L_v(x)$
◉	$P_v(x)$
○	$B_v(x)$
○	$J_v(x)$

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The linear normal form of

$$2\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 5y = 0$$

by using

$$y = x_1, y' = x_1' \text{ and } y'' = x_2'$$

Select the correct option

○	$x_1' = x_2, x_2' = -2x_2 + \frac{5}{2}x_1$
○	$x_1' = x_2, x_2' = 2x_2 + \frac{5}{2}x_1$
○	$x_1' = x_2, x_2' = -2x_2 - \frac{5}{2}x_1$
○	$x_1' = x_2, x_2' = 2x_2 - \frac{5}{2}x_1$

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Question # 7 of 10 (Start time: 12:53:40 AM, 01 February 2023) Total Marks: 1

$$\frac{dA}{dt} = \left[\frac{da_{ij}}{dt}\right]_{m\times n} \text{ is the } _____ \text{ of a matrix of a function.}$$

Select the correct option

Reload Math Equations

- row
- column
- integral
- derivative

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Click to Save Answer & Move to Next Question

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Question # 9 of 10 (Start time: 03:58:02 PM, 03 February 2023)

This equation

$$\det(A - \lambda I)$$

is known as _____of the matrix A.

Select the correct option

Rel

○	Linear equation
○	Algebraic equation
◉	Characteristic equation
○	All of these

Click to Save Answer & Mov

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The Legendre polynomials can also be generated by Rodrigues formula, $P_n(x) = \frac{1}{2^n\, n!}\frac{d^n}{dx^n}(x^2-1)^n$.

rrect option

Reload Math Equ

False

True



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Question # 8 of 10 (Start time: 03:58:26 PM, 03 February 2023)

Total Marks:

The regular singular point of the differential equation

$$(x^2 - 4)^2 y + (x - 2)y' + y = 0$$

is _____

Select the correct option

Reload Math Equations

- 2
- -2
- -1
- 1

Click to Save Answer & Move to Next Question

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estion # 7 of 10 (Start time: 03:49:28 PM, 03 February 2023)

Vector $\begin{pmatrix} -1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ are linearly . . . . . .

ect the correct option

- Dependent
- Independent

Click t

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Question # 10 of 10 (Start time: 03:59:27 PM, 03 February 2023)

If

$$X_1 = \begin{pmatrix} e^{-2t} \\ -e^{-2t} \end{pmatrix}$$

then

$$\frac{d}{dt}(X_1)$$

= _____

Select the correct option

○	$\begin{pmatrix} -2e^{2t} \\ 2e^{2t} \end{pmatrix}$
○	$\begin{pmatrix} -2e^{2t} \\ -2e^{2t} \end{pmatrix}$
○	$\begin{pmatrix} 2e^{2t} \\ 2e^{2t} \end{pmatrix}$
○	$\begin{pmatrix} 2e^{2t} \\ -2e^{2t} \end{pmatrix}$

Click to Save Answe

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Question # 9 of 10 (start time: 04:19:21 PM, 03 February 2023)

Total Marks:

The equation, $(x^2-4)^2\frac{d^2y}{dx^2}+(x-2)\frac{dy}{dx}+y=0$, has the singular point(s) at $x=$ _____.

Select the correct option

Reload Math Equations

○	± 2
○	$\pm i$
○	± 1
○	0

Click to Save Answer & Move to Next Question

Scanned with CamScanner

The Differential Equation

$$(x^2 - 16)y'' + 2xy' + 16y = 0$$

has singularity at

Select the correct option

None of these

$x = -16$

$x = \pm 4$

$x = 16$

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The equation, $\det(A - \lambda I) = 0$, is known as the characteristic equation.

True



False

Click to Save Answer & Mo

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Question #10 of 10 (Start time: 04:20:16 PM, 03 February 2023) Total Marks:

The matrix form of non - homogeneous system of differential equations is

$$X' = -3x + 4y - 9z$$
$$X' = 6x - y$$
$$X' = 10x - 4y + 3z$$

Select the correct option

Reload Math Equations

- $X' = \begin{bmatrix} -3 & -9 & -4 \\ 6 & 0 & -1 \\ 10 & 3 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$
- $X' = \begin{bmatrix} -3 & 4 & x \\ 6 & -1 & y \\ 10 & -4 & z \end{bmatrix} \begin{bmatrix} -9 \\ 0 \\ 3 \end{bmatrix}$
- $X' = \begin{bmatrix} -3 & 4 & -9 \\ 6 & -1 & 0 \\ 10 & -4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$
- $X' = \begin{bmatrix} -3 & 4 & -9 \\ 0 & 6 & 0 \\ -4 & 10 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Click to Save Answer & Move to Next Question

Scanned with CamScanner

The non - trivial solution of the system of differential equation exist only when

correct option

Reload Math Eq

$$del(X - \lambda I) = 0$$

$$del(A - \lambda I) \neq 0$$

$$del(A - \lambda I) = 0$$



$$del(A + \lambda I) = 0$$

Click to Save Answer & Move to Next Qu

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The simultaneous ordinary differential equations involve two or more equations that contain derivatives of two or more unknown functions of _____ independent variable

Select the correct option

Reload Math Equations

○	2
○	1
○	2
○	infinite

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The given system without the use of matrices

$$\frac{d}{dt}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 5 & 7 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix}e^t$$

is _____.

elect the correct option

Reload M

- $\frac{dx}{dt} = 2x + 4y + e^t; \frac{dy}{dt} = 5x + 7y + e^t$
- $\frac{dx}{dt} = 2x + 4y + e^{-t}; \frac{dy}{dt} = 5x + 7y + e^{-t}$
- $\frac{dx}{dt} = 2x + 4y - e^t; \frac{dy}{dt} = 5x + 7y - e^t$
- $\frac{dx}{dt} = 2x + 4y + e^t; \frac{dy}{dt} = 5x + 7y - e^t$

Click to Save Answer & Move to N

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The Differential Equation

$$(x^2 - 16)y'' + 2xy' + 16y = 0$$

has singularity at

Select the correct option

Reload Math Equations

○	$x = 16$
○	None of these.
○	$x = -16$
○	$x = \pm 4$

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A solution of a system of differential equations is a set of differentiable functions, $x = f(t)$, $y = g(t)$, $x = h(t), \ldots,$ those satisfy _____ equation(s) of the system on some interval I.

rrect option

Reload Math E

each

two

one

at least one

Click to Save Answer & Move to Next Q

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The linear normal form of

$$4\frac{d^3y}{dt^3} + y = e^t$$

,by using

$$y = x_1, y' = x_1', y'' = x_2' \text{and } y''' = x_3'$$

, is_____

Select the correct option

○	$x_1' = x_2, x_2' = x_3, x_3' = \frac{1}{4}x_1 + \frac{1}{4}e^t$
○	$x_1' = x_2, x_2' = x_3, x_3' = \frac{1}{4}x_1 - \frac{1}{4}e^t$
○	$= x_2, x_2' = x_3, x_3' = \frac{-1}{4}x_1 + \frac{1}{4}e^t$
○	$x_1' = x_2, x_2' = x_3, x_3' = -\frac{1}{4}x_1 - \frac{1}{4}e^t$

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Question # 10 of 10 (Start time: 04:00:11 PM, 03 February 2023) **Total Marks:**

The irregular singular point of the differential equation

$$(x^2-4)^2y+(x-2)y'+y=0$$

is _____.

Select the correct option

Reload Math Equations

○	-1
○	2
○	1
○	-2

Click to Save Answer & Move to Next Question

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Method of Frobenius

To solve a differential equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ about a regular singular point we employ the Frobenius' Theorem.

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Question # 8 of 10 (Start time: 03:58:03 PM, 03 February 2023)

The multiplicative identity of a 3×3 matrix is

Select the correct option

- $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
- $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

Clic

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A singular point

$$x = x_0$$

of the differential equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$$

is said to be a _____ point if both

$$(x - x_0)P(x)$$

and

$$(x - x_0)^2 Q(x)$$

are analytical at

$$x_0$$

.

Select the correct option

○	none of them
○	ordinary
○	regular singular
○	irregular singular

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(Quiz#3) Quiz Start Time: 03:44 PM, 03 Febru

:50:31 PM, 03 February 2023) Tota

Legendres polynomials are specific _____ degree polynomials.

Reload Math Equ



n

$n-1$

$n+1$

n^2

Click to Save Answer & Move to Next Que

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uestion # 8 of 10 (Start time: 05:20:50 PM, 03 February 2023)

An immediate consequence of the principle of superposition is that a constant multiple of any solution vec
order differential equation is also a solution of the system.

lect the correct option

- False
- True



Click to Sav

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Recurrence relation for polynomials relates

Select the correct option

○	Polynomials of different degrees.
○	Equal polynomials.
○	Polynomials of same degrees.
	All options are true.

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estion # 8 of 10 (Start time: 12:55:23 AM, 01 February 2023) **Total Mar**

Operator method or systematic elimination method of solution of a system of linear homogeneous or linear non-homogeneous differential equations is based on the process of systematic elimination of the _____ variables.

lect the correct option

- dependent
- independent

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Question # 6 of 10 (Start time: 06:41:32 PM, 01 February 2023) **Total Marks: 1**

This equation

$$\det(A - \lambda I)$$

is known as _____of the matrix A.

Select the correct option

Reload Math Equations

- ○ All of these
- ○ Algebraic equation
- ◉ Characteristic equation
- ○ Linear equation

Activate Windows

Click to Save Answer & Move to Next Question

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uestion # 9 of 10 (Start time: 06:43:50 PM, 01 February 2023) Total Marks: 1

Vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are linearly – – – – –.

lect the correct option

Reload Math Equations

- Dependent
- Independent



Activate Windows

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Question # 8 of 10 (Start time: 06:42:59 PM, 01 February 2023) **Total Marks: 1**

The irregular singular point of the differential equation

$$(x^2-4)^2y+(x-2)y'+y=0$$

is _____.

Select the correct option

Reload Math Equations

○	-1
○	1
◉	-2
○	2

Activate Windows

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Question # 4 of 10 (Start time: 06:40:07 PM, 01 February 2023) **Total Marks: 1**

The multiplicative identity of a 3×3 matrix is

Select the correct option

Reload Math Equations

- $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$
- $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$
- $\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$

Activate Windows
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Question # 2 of 10 (Start time: 12:47:14 AM, 01 February 2023) Total Marks:

The horizontal lines in a matrix are called the _____ of the matrix.

Select the correct option

- rows
- columns

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pherical Bessel functions are used in many physical problems.

ct the correct option

False

True



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Click to Sc e to Next Question

BC200404039: MUHAMMAD SADDIQUE

Time Left 8

MTH401 - Differential Equations (Quiz#3)

Quiz Start Time: 11:44 AM, 03 February 2

Question # 2 of 10 (Start time: 11:45:46 AM, 03 February 2023)

Total Mar

Legendres polynomials are specific _____ degree polynomials.

Select the correct option

Reload Math Equatio

- ○ $n-1$
- ○ n^2
- ○ $n+1$
- ○ n

Click to Save Answer & Move to Next Question

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Question # 9 of 10 (Start time: 12:56:38 AM, 01 February 2023) **Total Marks:** 1

If n is positive integer, then the solution of Legendre's differential equation is called a Legendre's _____ of degree n.

Select the correct option

Reload Math Equations

- function
- polynomial

Activate Windows

Question # 1 of 10 (start time: 03:50:34 PM, 03 February 2023)

The method of Frobenius always provide two solutions

Select the correct option

- True
- False

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Question # 1 of 10 (Start time: 12:45:41 AM, 01 February 2023) **Total Marks:** 1

A second order linear differential equation of the form

$$(1-x^2)y-2xy'+n(n+1)y=0$$

is called _____ differential equation.

Select the correct option

Reload Math Equations

- ○ Picard
- ○ Bernoulli
- ◉ Legendre
- ○ Bessel

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uestion # 10 of 10 (Start time: 12:57:16 AM, 01 February 2023) **Total Marks: 1**

To solve a differential equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$$

about a regular singular point we employ the _____ theorem.

lect the correct option

Reload Math Equations

- none of them
- Frobenius
- Bessel
- Legendre

Activate Windows

The linear normal form of

$$4\frac{d^3y}{dt^3} + y = e^t$$

,by using

$$y = x_1, y' = x_1', y'' = x_2' \text{and } y''' = x_3'$$

, is_____.

Select the correct option

Reload Math Equations

○	$x_1' = x_2, x_2' = x_3, x_3' = -\frac{1}{4}x_1 - \frac{1}{4}e^t$
◉	$x_1' = x_2, x_2' = x_3, x_3' = \frac{1}{4}x_1 + \frac{1}{4}e^t$
○	$x_1' = x_2, x_2' = x_3, x_3' = \frac{1}{4}x_1 - \frac{1}{4}e^t$
○	$x_1' = x_2, x_2' = x_3, x_3' = \frac{-1}{4}x_1 + \frac{1}{4}e^t$

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MTH401 - Differential Equations (Quiz#3)

Question # 1 of 10 (Start time: 11:44:09 AM, 03 February 2023)

The solution of Legendre differential equation is denoted by _____.

Select the correct option

○	$L_v(x)$
○	$B_v(x)$
○	$J_v(x)$
○	$P_v(x)$

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Question # 2 of 10 (Start time: 06:38:12 PM, 01 February 2023) Total Marks: 1

The Legendre polynomials can also be generated by Rodrigues formula, $P_n(x) = \frac{1}{2^n\, n!}\frac{d^n}{dx^n}(x^2-1)^n$.

Select the correct option

Reload Math Equations

○	False
◉	True

Activate Windows

Click to Save Answer & Move to Next Question

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Select the correct option

- (●) True
- ( ) False



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Go to Settings to activate Windows.

Click to Save Answer & Move to Next Question

stion # 7 of 10 (Start time: 10:33:09 AM, 03 February 2023) Total Ma

olution of Legendre's differential equation is called

ct the correct option

- Recurrence relation
- Legendre function
- Legendre polynomial
- Legendre differential



Click to Save Answer & Move to Next Questio

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1401 - Differential Equations (Quiz#3)

Quiz Start Time: 10:25 AM, 03 February

stion # 5 of 10 (Start time: 10:31:22 AM, 03 February 2023)

Total M

he simultaneous ordinary differential equations involve two or more equations that contain derivatives of two or more unknown functions
_____ independent variable

ct the correct option

Reload Math Equati

2

infinite

2

1



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MTH401 - Differential Equations (Quiz#3)

Question # 1 of 10 (Start time: 11:44:09 AM, 03 February 2023)

The solution of Legendre differential equation is denoted by _____.

Select the correct option

○	$L_v(x)$
○	$B_v(x)$
○	$J_v(x)$
○	$P_v(x)$

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The equation, $(x^2-4)^2\frac{d^2y}{dx^2}+(x-2)\frac{dy}{dx}+y=0$, has the singular point(s) at $x=$ ____

± 1

0

$\pm i$



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Lecture 34

Legendre's Differential Equation

A second order linear differential equation of the form

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

is called Legendre's differential equation and any of its solution is called Legendre's function. If n is positive integer then the solution of Legendre's differential equation is called a Legendere's polynomial of degree n and is denoted by $P_n(x)$

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) solution of Bessel equation is denoted by _____

ct the correct option

$P_n(x)$

$J_n(x)$

$B_n(x)$

$L_n(x)$

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